R-Slip Robot Experiment Automation - Introduction Summary

Project Overview

The R-Slip is a hexapod robot platform designed for slip-based locomotion research. This project automates the complex multi-machine setup process, reducing 23+ manual SSH commands across distributed systems (Windows, sbRIO controller, Jetson Orin Nano boards) into a single-click launcher.

Problem Solved

Before: Operators required:

- Manual SSH connections to multiple machines
- Sequential setup of network configurations
- Starting and verifying multiple services
- Coordinating three parallel system activations
- High error-prone complexity with 23+ individual steps

After: One-click automation handling:

- Intelligent IP detection and validation
- Automatic multi-window orchestration
- Service startup and health verification
- Complete environment variable configuration
- User-guided setup with clear feedback

What It Does

1. **Orchestrates Multi-Machine Setup**
    - Windows PC → sbRIO (controller)
    - Windows PC → Jetson Orin Nano (Orin1: ROS bridge)
    - Windows PC → Jetson Orin Nano (Orin2: control)
2. **Automates Service Startup**
    - SSH authentication
    - grpccore service initialization
    - fpga_driver service activation
    - Network port verification (50051)
3. **Configures ROS2 Environment**
    - Sources ROS2 Humble setup
    - Sets master address and local IP
    - Starts communication bridge
    - Prepares control interface

4. **Enables Robot Operation**
    - One-click power sequence
    - Sensor calibration
    - Standing position control
    - Tripod gait execution

Key Features

- ☑ **Single-Click Launch** - One batch file handles everything
- ☑ **Intelligent IP Detection** - Auto-finds sbRIO, manual entry for customization
- ☑ **Multi-Window Coordination** - Manages 3 parallel SSH sessions
- ☑ **Service Verification** - Confirms all systems operational
- ☑ **Comprehensive Documentation** - Setup guide, troubleshooting, quick reference
- ☑ **Pre-Configured Commands** - Ready-to-execute robot control sequence

Technical Stack

- **Automation:** Batch scripts (.bat) and PowerShell (.ps1)
- **Connectivity:** SSH over WiFi
- **Platforms:** Windows 10/11, NI sbRIO, Jetson Orin Nano
- **Middleware:** ROS2 Humble, gRPC
- **Network:** 192.168.0.x WiFi, port 50051

Workflow

1. Run `RSlip_Robot_Setup.bat`
2. Input sbRIO IP (auto-detected or manual)
3. Wait 30 seconds for 3 windows to open
4. Verify services running (sbRIO, Orin1, Orin2)
5. Execute robot control commands in sequence
6. Start tripod gait experiment

Impact

- **Reduces Setup Time:** 15+ minutes → 30 seconds
- **Eliminates Manual Errors:** Scripted automation vs manual typing
- **Improves Reproducibility:** Consistent startup procedure
- **Lowers Learning Curve:** Single script vs 23+ individual commands
- **Saves Developer Time:** Focus on research, not system setup

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Files Delivered:

- RSlip_Robot_Setup.bat (main launcher)
- 3 supporting PowerShell/Batch scripts
- SETUP_GUIDE.md (comprehensive documentation)

Status: ☑ Complete and ready for deployment